

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q18-0919

Rustburg 138 kV
December 17, 2019

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q18-0919. The Interconnection Customer proposes a 22.5 MW battery energy storage facility interconnecting at the Rustburg 138 kV substation in the Virginia area. The request was received on January 13, 2018. The study evaluated system intact and single contingency conditions using the regional planning case, including 271 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 4 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$875,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
SCOTTSVILLE - REACH 138 KV L	BLUFF CITY - REACH 138 KV	117.2	256	9.2
TREGO - BEAVER 500 KV	ADAMS - ONANCOCK DIESEL GENERATION FACILITY 230 KV	104.5	5,059	8.5
TREGO - TROY ENERGY LLC 500 KV	TREGO - SWAN VLY 500 KV	103.6	4,121	8.2
BEARDSLEY ST - RED HILLS 138 KV Line	HINTON AP SUBSTATION - LOWMOOR 345 KV Line	103.6	355	6.4

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
SCOTTSVILLE - REACH 138 KV L	Rebuild line with higher-capacity conductor	\$8,660,000	1% (\$109,000)
TREGO - BEAVER 500 KV	Reconductor line	\$125,195,000	0% (\$422,000)
TREGO - TROY ENERGY LLC 500 KV	Reconductor line	\$92,113,000	0% (\$313,000)
BEARDSLEY ST - RED HILLS 138 KV Line	Reconductor line	\$4,302,000	1% (\$31,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
	Total allocated to project		\$875,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.